

Seminar on Algebraic Geometry of Bois Marie

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Intersection Theory and the Riemann–Roch Theorem

(SGA 6)

a seminar directed by

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Preface

The present edition of *Séminaire de Géométrie Algébrique 6* reproduces, up to minor changes, the first edition, distributed by the I.H.E.S. in the form of mimeographed exposés. The corrections made concern essentially printing errors, except in Exposés I and III, where slight modifications were made by L. Illusie to a few statements which were incorrect in the primitive version. Footnotes have also been added, especially in Exposé XIV by A. Grothendieck, in order to point out recent progress on questions that were still open at the time of the seminar.

Finally, note that Exposé XI, by D. Ferrand, on determinant theory for perfect complexes, was not written up and therefore will not be published; to avoid errors in references, we have left unchanged the numbering of the following exposés.

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Introduction

The purpose of the present seminar is to develop a global theory of intersections, and a Riemann–Roch formula, for arbitrary schemes. The reader will find a description of the programme of the seminar in Exposé 0. The possibility in principle of proving a Riemann–Roch formula for general regular schemes, along the lines of the well-known Borel–Serre report of 1958, was already clear at that time, at least for a projective morphism. The extension to the general case is less evident; the programme in which it belongs, and which is partly carried out in this seminar, goes back to 1960. As was also the case for the duality theory of coherent sheaves, an essential tool for formulating a satisfactory theory is Verdier’s theory of derived categories; familiarity with it is indispensable for understanding the seminar.

Apart from this theory, the study of the seminar requires little more than a general knowledge of the foundations of algebraic geometry as set out in EGA, chapters I, II, III. We shall only occasionally refer to EGA IV, for certain facts concerning flat or smooth morphisms, which for the most part also appear in the first exposés of SGA 1. Finally, to develop some results with the full generality desirable for applications, we sometimes use the notion of a ringed site and of a ringed topos, for which we refer to SGA 4, Exposés I to IV. A reader unfamiliar with the language of sites and topoi may everywhere replace these objects by ordinary topological spaces, the objects of the topos then being replaced by open subsets of these spaces; nevertheless,

we recommend that the reader assimilate the language of topoi, which provides an extremely convenient unifying principle.

The fundamental notion for the theory presented here is that of a perfect complex of modules, together with its various variants, developed in Exposés I, II, III. It seems clear that these notions, also important in other contexts in algebraic geometry, notably for formulas of Lefschetz–Verdier type in cohomology with discrete coefficients (SGA 5) or coherent coefficients, will also have a role to play outside algebraic geometry, especially in the formulation of a complex analytic variant of the Riemann–Roch theorem presented here, or of suitable variants of the Atiyah–Singer index theorem. Some indications in this direction will be given in Exposé II. Unfortunately, at present there is still no statement, even heuristic, which would encompass these two theorems, the first of which remains conjectural for the moment. A new idea seems to be missing, just as one is also missing in algebraic geometry for proving Riemann–Roch beyond projective hypotheses (cf. Exposé XIV, no. 2).

We shall make no use in this seminar of the local theory of intersections, set out in J.-P. Serre, *Algèbre locale. Multiplicités* (Lecture Notes in Mathematics no. 11, Springer), and merely point out here that Serre’s course had an evident influence on the genesis of the theory in 1957. Nor shall we use the theory of the Chow ring developed in the Chevalley Seminar of 1958, Exposés 2 and 3 (École Normale Supérieure). That theory is essentially tied to nonsingularity hypotheses, whereas the aim of our seminar is, on the contrary, to develop a theory of intersections on general schemes, and even on general locally ringed topoi. On this matter see the comments in Exposé XIV, nos. 4 and 8, which give the relations between our theory and that of the Chow ring and raise the question of a generalization of the latter. Thus one may say that the seminar presents a coherent and self-contained theory of intersections, but one which is by no means exhaustive of the different points of view useful, indeed indispensable, in intersection theory; consequently it should be complemented by the cited exposés of Serre and Chevalley.

We have appended to the seminar, as an appendix to Exposé 0, Grothendieck’s 1957 report, “Classes of sheaves and the Riemann–Roch theorem”, which served as the basis for the Borel–Serre seminar at Princeton in the same year and for their already cited article. That report sketches two proofs of the Riemann–Roch theorem for nonsingular quasi-projective varieties. The first, valid for the time being only in characteristic zero but giving, in return, a slightly more precise result in the case of an immersion, has not yet appeared in the literature, apart from the work of Borel–Serre and the present seminar. The report presupposes, of course, none of the other exposés of the seminar, and can even serve, like Exposé 0, as an introduction to the study of the latter, especially for a reader not yet familiar with Borel–Serre. The proof just alluded to applies essentially unchanged to the case of a projective morphism of complex analytic spaces, and will doubtless also apply in arbitrary characteristic once the problem of the operations “exterior powers” in the derived category is solved (cf. Exposé XIV, nos. 1, 2, 3). The absence of a study of this question is no doubt the most serious gap in this seminar, from the point of view we have adopted.

One will notice the absence, in the table of contents of the present seminar, of two of the participants mentioned on the cover page. One of them, J.-P. Jouanolou, took an active part in the technical elaboration of the first part of the seminar, but was prevented from taking part in the oral exposés; in particular, he is responsible for the linear-independence assertion contained in the important statement VI 1.1, giving the structure of the ring K° of classes of locally free modules on a projective bundle, which had previously been proved only under the assumption that the base scheme has an ample invertible module. The other, J.-P. Serre, gave two oral exposés which were logically independent of the rest of the seminar, and therefore preferred to publish them in the form of a separate article (cf. J.-P. Serre, “Grothendieck groups of split reductive group schemes”, to appear in *Publications Mathématiques*, no. 34).

As in the other parts of the *Séminaire de Géométrie Algébrique du Bois Marie*, the sigla SGA 1 to SGA 6 refer to the different parts of that seminar; the Roman numeral following the siglum indicates the number of the exposé, and the Arabic numerals following it correspond to the internal numbering of the exposé in question. For internal references within the present seminar the siglum SGA 6 is omitted, and references of the form I 4.9 are used, meaning: Exposé I, statement 4.9. The siglum EGA refers to the *Éléments de Géométrie Algébrique* of J. Dieudonné and A. Grothendieck.

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Exposé 0

Outline of a programme for a theory of intersections on general schemes

by A. Grothendieck

The present exposé is introductory in nature, and its reading is not logically indispensable for the study of the seminar. It is addressed more particularly to readers familiar with the provisional version of the Riemann–Roch theorem, as it is presented in the Borel–Serre report or in Grothendieck’s report already mentioned in the introduction, cited below as [RRR], and reproduced as an appendix at the end of the present exposé.

1. Reminder on the Riemann–Roch formula

Recall the Riemann–Roch formula for a proper morphism

$$f : X \longrightarrow Y$$

of quasi-projective smooth schemes over a field k , and a coherent sheaf F on X , whose class in the group $K(X)$ of classes of coherent sheaves on X is denoted by $\mathrm{cl}(F)$:

$$\mathrm{Todd}(T_Y) \, \mathrm{ch}_Y(f_!(\mathrm{cl}(F))) = f_*(\mathrm{Todd}(T_X) \, \mathrm{ch}_X(F)). \quad (1.1)$$

This formula is valid in $A(Y) \otimes \mathbb{Q}$, where $A(Y)$ is the Chow ring of Y . The f_* on the right hand side is obtained by tensoring with $\mathbb{Q}$ the homomorphism “direct image of cycles”

$$f_* : A(X) \longrightarrow A(Y),$$

while the term on the left is the Euler–Poincaré characteristic of F relative to f :

$$f_!(\mathrm{cl}(F)) = \sum_i (-1)^i \mathrm{cl}(R^i f_*(F)).$$

The symbols ch_X , ch_Y denote the Chern characters on X and on Y , and T_X , T_Y the tangent bundles of X and Y . As is known, $\mathrm{Todd}(-)$ and $\mathrm{ch}(-)$ are universal polynomials with coefficients in $\mathbb{Q}$ in the Chern classes of the argument. Since the constant term of $\mathrm{Todd}(-)$ is 1, it is an invertible element for every value of the argument. Thus, after multiplying by $\mathrm{Todd}(T_Y)^{-1}$, formula (1.1) takes the equivalent form, more convenient for our purposes,

$$\mathrm{ch}_Y(f_!(\mathrm{cl}(F))) = f_*(\mathrm{Todd}(T_f) \, \mathrm{ch}_X(F)), \quad (1.2)$$

where

$$T_f = T_X - f^* T_Y \in K(X). \quad (1.3)$$

Thus T_f plays the role of a virtual relative tangent bundle of X over Y . When f is smooth, one simply has $T_f = T_{X/Y}$, whereas when $f : X \rightarrow Y$ is an immersion, one obtains

$$T_f = -\check{N}_{X/Y},$$

where $N_{X/Y}$ denotes the normal sheaf of X in Y .

One of the main aims of the seminar is to generalize (1.2) simultaneously in two directions:

- a) to get rid of the hypothesis that a base field k exists;
- b) to replace the regularity hypotheses on Y, X by a hypothesis of “local regularity” on f .

Finally, along the way, we shall also deal with the following problem:

- c) to eliminate the quasi-projectivity hypotheses which, in the absence of a base field, are expressed by the existence of ample invertible modules on X and Y .

2. Removing the base field

Let us first examine generalization a), while retaining the hypotheses of regularity and of the existence of ample invertible modules on X and on Y .

The definitions of $K(X)$, $K(Y)$ and of the homomorphism $f_! : K(X) \rightarrow K(Y)$ then present no new problem, thanks to the fact that X and Y are regular. The most natural way to give

a meaning to (1.2) would seem to consist in defining Chow rings $A(X), A(Y)$ and a group homomorphism

$$f_* : A(X) \longrightarrow A(Y),$$

in establishing a theory of Chern classes, giving maps

$$c_i : K(X) \longrightarrow A(X),$$

and similarly for Y , and finally in describing an element, the virtual relative tangent bundle,

$$T_f \in K(X).$$

2.1

For the latter, a definition presents itself rather evidently. One uses the fact that the morphism f , thanks to the hypothesis that ample invertible modules exist on X and on Y , may be factored as

$$X \xrightarrow{i} X' \xrightarrow{f'} Y,$$

where i is a closed immersion and f' is smooth; for example, one may take $X' = \mathbb{P}_Y^r$ for suitable r . One then sets

$$T_f = i^* T_{X'/Y} - \check{N}_{X/X'} \in K(X), \quad (2.1)$$

where $\Omega_{X'/Y}^1$ is the locally free module of relative 1-differentials of X' over Y , or the relative cotangent module of X' over Y , and $N_{X/X'}$ is the conormal module of X in X' , by definition equal to J/J^2 , where J is the ideal on X' defining X . The regularity hypotheses on X and X' also imply that $N_{X/X'}$ is locally free. Finally, $\check{N}$ denotes the dual of N . One verifies without difficulty, in Exposé VIII, that the element T_f defined by (2.1) is independent of the chosen factorization of f .

2.2

As for a definition, under the conditions considered, of a Chow ring and a corresponding theory of Chern classes, although it is plausible that such a theory should be developable on the model of smooth varieties over a field, it seems that this work has not been done at the present time. We shall therefore adopt another viewpoint, by directly defining a ring which replaces the Chow ring, and whose tensor product with $\mathbb{Q}$ is indeed isomorphic to $A(X) \otimes \mathbb{Q}$ in the classical case.

A first natural filtration of $K(X)$, the one considered in [RRR], is the topological filtration by the codimension of the support of coherent sheaves: $\text{Fil}^i K(X)$ is generated by the $\text{cl}(F)$ with F coherent and $\text{codim}(\text{Supp } F, X) \geq i$. A first question is the compatibility of this filtration with the multiplicative structure of $K(X)$. In [RRR], this compatibility is obtained, for X smooth over a field, by means of Chow's moving lemma. Assuming this assertion, one obtains a graded ring $\text{Gr}_{\text{top}}(X)$ and a canonical surjective homomorphism

$$\varphi : A(X) \longrightarrow \text{Gr}_{\text{top}}(X),$$

sending the class of a prime cycle Z to $\text{cl}(\mathcal{O}_Z)$; it then turns out, Exposé XIV 4.2, that the kernel is a torsion subgroup. Moreover, [RRR] shows how the images in $\text{Gr}_{\text{top}}(X)$ of the Chern classes $c_i(F)$ can be computed directly in terms of $\text{cl}(F) \in K(X)$ and the exterior-power operations λ^i in $K(X)$.

This shows that, for a noetherian regular scheme X endowed with an ample invertible module, the ring $\text{Gr}_{\text{top}}(X)$ is a convenient substitute for the hypothetical Chow ring $A(X)$. It also has

the advantage over the latter that certain computations of [RRR], leading to important formulas in this ring, are performed directly in it—formulas which, even for X projective and smooth over a field, have not at present been established in $A(X)$ itself (Exp. XIV, 4.3 and 6.4). To justify this viewpoint completely, however, one would have to solve the problem mentioned above of the compatibility of the topological filtration of $K(X)$ with its ring structure, a problem which seems essentially connected with Chow's moving lemma, itself the essential technical ingredient of the theory of the Chow ring which we have claimed to be able to avoid.

2.3

To avoid this problem, one endows $K(X)$ with a second filtration, finer than the topological filtration, and describable in purely algebraic terms from the structure of $K(X)$ as an augmented λ -ring. The definition of this filtration is suggested by the expression given in [RRR] for the Chern classes of an element $x \in K(X)$ in $\mathrm{Gr}_{\mathrm{top}}(X)$:

$$c_i(x) = \gamma^i(x - \epsilon(x)) \mod \mathrm{Fil}^{i+1} K(X), \quad (2.3)$$

where γ^i denotes a certain linear combination of the λ^j ($j \leq i$), made explicit in loc. cit., and where ϵ is the augmentation. One shows that $\gamma^i(x - \epsilon(x))$ is indeed of topological filtration at least i , which gives meaning to formula (2.3). One then defines the λ -filtration of $K(X)$ by the ideals $\mathrm{Fil}_\lambda^i K(X)$, where $\mathrm{Fil}_\lambda^i K(X)$ consists of finite sums of expressions of the form

$$\gamma^{i_1}(x_1 - \epsilon(x_1)) \cdots \gamma^{i_r}(x_r - \epsilon(x_r)), \quad i_1 + \cdots + i_r \geq i.$$

The associated graded ring for this filtration will be denoted $\mathrm{Gr}^\bullet(X)$; it is this ring which will replace the Chow ring of X in the seminar. In the case under consideration here, one can moreover show that the canonical homomorphism

$$\mathrm{Gr}^\bullet(X) \longrightarrow \mathrm{Gr}_{\mathrm{top}}(X) \quad (2.4)$$

is an isomorphism modulo torsion, justifying the viewpoint that, after tensoring with $\mathbb{Q}$, this ring can play exactly the same role as the Chow ring tensored with $\mathbb{Q}$. On the other hand, exactly what was needed has been done so that formula (2.3), with $\mathrm{Fil}_{\mathrm{top}}$ replaced by Fil_λ , defines a theory of Chern classes with values in $\mathrm{Gr}^\bullet(X)$; the Chern classes in $\mathrm{Gr}_{\mathrm{top}}(X)$ are then deduced by means of the homomorphism (2.4). It is shown in Exposé V, using the fact proved in Exposé VI that $K(X)$ is a special λ -ring in the sense of [RRR], that is, a λ -ring in the sense of Exposé V, that this notion of Chern class satisfies all the usual formal properties reviewed in [RRR].

2.4

One should note a very appreciable advantage of the definition of $\mathrm{Gr}^\bullet(X)$ over that of $A(X)$ and $\mathrm{Gr}_{\mathrm{top}}(X)$: it remains meaningful for an absolutely arbitrary scheme X , and more generally for an arbitrary locally ringed topos X . One then takes for $K(X)$ the group of classes constructed from locally free modules on X , which is again an augmented λ -ring, hence equipped with a corresponding filtration and an associated graded $\mathrm{Gr}^\bullet(X)$. The price of this generality, however, is that when one is not willing to neglect torsion phenomena, i.e. to tensor with $\mathbb{Q}$, the properties of $\mathrm{Gr}^\bullet(X)$ are in several respects less satisfactory than those of its two competitors (Exp. XIV no. 4). Thus, assuming again that $f : X \rightarrow Y$ is a proper morphism of noetherian regular schemes endowed with ample invertible modules, it is not true in general that the homomorphism

$$f_* : K(X) \longrightarrow K(Y)$$

maps $\mathrm{Fil}_\lambda^i K(X)$ into $\mathrm{Fil}_\lambda^{i-d} K(Y)$, where d is the relative dimension of X over Y , and hence defines, by passage to associated graded, a degree $-d$ homomorphism from $\mathrm{Gr}^\bullet(X)$ to $\mathrm{Gr}^\bullet(Y)$. However, Exposé VII proves that this is true after tensoring with $\mathbb{Q}$, so that one obtains a degree $-d$ homomorphism

$$f_* : \mathrm{Gr}^\bullet(X) \otimes \mathbb{Q} \longrightarrow \mathrm{Gr}^\bullet(Y) \otimes \mathbb{Q}, \quad (2.5)$$

which plays the role of the homomorphism “direct image of cycles” $A(X) \rightarrow A(Y)$.

Having said this, we see that we possess all the structural elements needed to give meaning to formula (1.2) when the existence of a base field is not postulated, with $f : X \rightarrow Y$ a proper morphism of noetherian regular schemes admitting ample invertible modules.

3. The case of general schemes

We now show how one can still give meaning to formula (1.2) when the regularity hypothesis on X, Y is abandoned and replaced by a local hypothesis on the morphism f , to be specified below.

3.1

A first difficulty comes from the fact that the observation which served as the starting point of the theory [RRR], namely that the group $K(X)$ of classes of sheaves on X is the same whether it is defined in terms of coherent sheaves or in terms of locally free sheaves, is essentially tied to the regularity hypothesis on X . For a general scheme X , say noetherian for simplicity, one must distinguish between these two groups; we shall denote them respectively by $K_\bullet(X)$ and $K^\bullet(X)$. The first is covariant in X for proper morphisms, the second contravariant in X for arbitrary morphisms. Notice that $K^\bullet(X)$ is naturally endowed with a ring structure, indeed with an augmented λ -ring structure, whereas $K_\bullet(X)$ no longer has, in general, a ring structure, but only a module structure over $K^\bullet(X)$. The homomorphism $x \mapsto x \cdot \mathrm{cl}(\mathcal{O}_X)$ from $K^\bullet(X)$ to $K_\bullet(X)$ is no longer an isomorphism in general. For our formulation of the generalized Riemann–Roch theorem, we shall work exclusively with $K^\bullet(X)$ and $K^\bullet(Y)$, to the exclusion of $K_\bullet(X)$ and $K_\bullet(Y)$. As noted in 2.3, these augmented λ -rings, equipped with their natural λ -filtration, give rise to graded rings $\mathrm{Gr}^\bullet(X)$, $\mathrm{Gr}^\bullet(Y)$, and to Chern-class homomorphisms

$$c_i : K^\bullet(X) \longrightarrow \mathrm{Gr}^\bullet(X),$$

and similarly for Y . The difficulty, however, lies in defining a direct-image homomorphism

$$f_* : K^\bullet(X) \longrightarrow K^\bullet(Y), \quad (3.1)$$

which can no longer be defined by the naive formula

$$f_*(\mathrm{cl}(F)) = \sum_i (-1)^i \mathrm{cl}(R^i f_*(F)),$$

because even if F is locally free, the modules $R^i f_*(F)$ are in general no longer locally free, nor even of finite Tor-dimension, and the preceding formula loses its meaning. A solution to this difficulty is furnished by the viewpoint of derived categories. Indeed, whereas the $R^i f_*(F)$ are in general “bad” sheaves, say of infinite Tor-dimension, the complex $Rf_*(F)$ on Y , the direct image of F in the sense of derived categories, tends to be excellent. More precisely, when F has finite Tor-dimension over Y , for instance when f has finite Tor-dimension and F is locally free on X , then $Rf_*(F)$ is perfect on Y , that is, has coherent cohomology and globally finite Tor-dimension (Exp. III). It follows easily, Exp. IV, that its class in the Grothendieck group of perfect complexes has a meaning in $K^\bullet(Y)$.

3.2

To give a meaning to (1.2), it remains to impose on f a hypothesis making it possible to give a meaning to the homomorphism (2.5), i.e. ensuring that the homomorphism (3.1) just defined is, up to a shift and modulo torsion, compatible with the filtrations of these rings; and finally one must again define an element, the “virtual relative tangent bundle”,

$$T_f \in K^\bullet(X).$$

For the latter, returning to the definition proposed in 2.1, one sees that a hypothesis is needed ensuring that, with the notation of loc. cit., the conormal module $N_{X/X'}$ is locally free. The most natural hypothesis to this end is that i be a “regular immersion”, i.e. identify X with a subscheme of X' defined by an ideal which locally can be defined by an $\mathcal{O}_{X'}$ -regular sequence of sections of $\mathcal{O}_{X'}$. This hypothesis, purely local in nature, in fact does not depend on the choice of the factorization of f into an immersion followed by a smooth morphism (Exp. VIII), and, as above, the same is true for the class T_f defined by formula (2.1). When the hypothesis just considered is satisfied, the morphism f is called a “locally complete intersection” morphism. Such a morphism is also locally of finite Tor-dimension.

We shall also allow ourselves to omit the word “locally” in this terminology, which cannot cause any confusion.